THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Sound cannot travel through: 1

(a) a vacuum **(b)** air **(c)** steel **(d)** water

2. The area under a velocity-time graph gives: 1

(a) displacement **(b)** speed **(c)** time **(d)** acceleration

3. Action and reaction do not cancel because they: 1

(a) act on different bodies **(b)** are unequal

(c) are in the same direction **(d)** act at different times

4. The law of conservation of energy states that energy: 1

(a) can neither be created nor destroyed but only changed from one form to another

(b) can be destroyed

(c) is always lost

(d) can be created

5. The pitch of a sound depends on its: 1

(a) medium **(b)** frequency **(c)** speed **(d)** amplitude

6. Acceleration due to gravity near the Earth's surface is about: 1

(a) 10 m/s squared **(b)** 10 m/s **(c)** 100 m/s squared **(d)** 1 m/s squared

7. Airbags in cars protect passengers by: 1

(a) stopping the car

(b) increasing the stopping time and reducing the force

(c) reducing the mass

(d) reducing the speed

8. Mechanical advantage = 1

(a) effort / load **(b)** load x effort **(c)** load - effort **(d)** load / effort

9. Bats find their prey in the dark by: 1

(a) echolocation using ultrasonic waves **(b)** infrasonic sounds

(c) light **(d)** smell

10. Displacement is a: 1

(a) scalar quantity **(b)** vector quantity **(c)** unit **(d)** constant

11. The SI unit of force is: 1

(a) joule **(b)** pascal **(c)** watt **(d)** newton

12. A girl carrying a box walks horizontally. The work done by the force holding the box is: 1

(a) negative **(b)** positive **(c)** maximum **(d)** zero

13. Sound waves in air are: 1

(a) longitudinal waves **(b)** light waves

(c) transverse waves **(d)** electromagnetic waves

14. A straight line parallel to the time axis on a distance-time graph indicates that the body is: 1

(a) falling **(b)** moving with uniform speed

(c) accelerating **(d)** at rest

15. A rocket moves upward because: 1

(a) friction

(b) the exhaust gases pushed downward exert an equal and opposite force on it

(c) the Earth attracts it

(d) air pushes it

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a passenger fall forward when a moving bus stops suddenly? 2

17. State the work-energy theorem. 2

18. Distinguish between pitch and loudness. 2

19. Why is uniform circular motion accelerated even though the speed is constant? 2

20. Why does a fielder draw his hands back while catching a fast ball? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the working of SONAR and calculate the depth of the sea if the echo of an ultrasound pulse is received after 4 s (speed in water = 1500 m/s). 3

22. The odometer of a car reads 2000 km at the start and 2050 km after 1 hour. Find the average speed and say what the speedometer shows. 3

23. Why do we need static friction to walk? What happens on ice? 3

24. A boy of mass 40 kg climbs 12 stairs of 20 cm each in 8 s. Find his power (g = 10). 3

OR

What is the effect of temperature on the speed of sound in air? Give a reason. 3

25. A car starts from rest with an acceleration of 3 m/s squared. Find its velocity after 4 s and the distance covered. 3

OR

A car of mass 1000 kg moving at 20 m/s is stopped in 5 s. Find the force applied by the brakes. 3

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SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain uniform circular motion with the tangent velocity direction, the relation v = 2 pi r/T, and calculate the speed of a body at the tip of a 0.5 m radius wheel with period 2 s (pi = 3.14). **5**

OR

A body of mass 5 kg is moving on a rough surface with an applied force of 30 N and friction of 10 N. Find the acceleration, the velocity after 4 s and the distance covered in 4 s starting from rest. **5**

27. Explain the conservation of energy in a swinging pendulum. Describe the changes of KE and PE and the effect of air resistance. **5**

OR

Explain reverberation, its causes and effects in an auditorium, and the methods of controlling it, with the principle of curved sound boards. **5**

END OF QUESTION PAPER — **Total: 50 Marks**

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